CHEST

Special Features

Statins and Interstitial Lung Disease*

A Systematic Review of the Literature and of Food and Drug Administration Adverse Event Reports

Antonio B. Fernández, MD; Richard H. Karas, MD, PhD; Alawi A. Alsheikh-Ali, MD; and Paul D. Thompson, MD

Objective: **To systematically review all published case reports and the US Food and Drug Administration adverse event reporting (FDA-AER) database to examine the relationship between statins and interstitial lung disease (ILD).**
Data sources: **PubMed (1987 to September 2007) and the FDA-AER database (as of June 2007) were searched for reports of ILD in which a statin was listed as a causative suspect.**
Review methods: **Two authors (one author for Pub Med cases and one for FDA-AER cases) independently abstracted patient data. Given the paucity of information, all case reports and case series in English and French were included. All adverse event reports from the FDA-AER database in which a statin was listed as causative suspect were included.**
Results: **The literature search using PubMed yielded eight articles describing a total of 14 case reports of ILD in association with statin use. The FDA-AER system database contained 162 cases of reported statin-induced ILD as of June 2007. For every 10,000 reports of a statin-associated adverse event, approximately 1 to 40 reports were for ILD.**
Conclusions: **Statin-induced ILD is a possible newly recognized side effect of statin therapy. The mechanism of lung injury is not defined. The current review provides novel information from the FDA-AER that supports a possible, although unusual, pulmonary class effect of statins.**
(CHEST 2008; 134:824–830)

Key words: drug-induced interstitial lung disease; hydroxymethylglutaryl coenzyme A reductase inhibitors; interstitial pneumonitis; parenchymal lung disease; pulmonary fibrosis; statins

Abbreviations: FDA-AER = Food and Drug Administration adverse event reporting; HMG-CoA = hydroxymethylglutaryl coenzyme A; ILD = interstitial lung disease

Since the introduction of lovastatin in 1987, hydroxymethylglutaryl coenzyme A (HMG-CoA) reductase inhibitors or statins have been proven to be a safe and effective treatment of hypercholesterolemia and to reduce cardiovascular events.[1,2] Myopathy and abnormalities in liver function test findings are the most commonly reported serious side effects of these drugs.[3] Interstitial lung disease (ILD) attributed to statin use was first described in 1995,[4] and there have been at least 13 additional reports of statin-associated lung disease.[4–7] The present systematic review examines the evidence

*From the Section of Cardiovascular Medicine (Dr. Fernández), Yale University School of Medicine, New Haven, CT; Division of Cardiology (Dr. Karas), Institute for Clinical Research and Health Policy Studies (Dr. Alsheikh-Ali), Tufts Medical Center, Tufts University School of Medicine, Boston, MA; and Division of Cardiology (Dr. Thompson), The Henry Low Heart Center, Hartford Hospital, Hartford, CT.
Dr. Thompson has received grant/research support from Merck, Pfizer, AstraZeneca, and B. Braun; is a consultant for AstraZeneca and Merck; is on the Speaker's Bureau for Merck, Pfizer, Abbott, AstraZeneca, and ScheringPlough; and owns stock in Schering Plough, Merck, Illumina, and Zoll. He has other financial or occasional speaking honoraria from Merck, Pfizer, Abbott, AstraZeneca, ScheringPlough, and Reliant. Dr. Karas is a consultant for Merck and Abbott, and has received honoraria from Merck and Abbott. Dr. Alsheikh-Ali is recipient of a faculty development award from Pfizer and Tufts Medical Center. Dr. Fernandez has no conflicts of interests to disclose.

Manuscript received April 23, 2008; revision accepted July 7, 2008.
Reproduction of this article is prohibited without written permission from the American College of Chest Physicians (www.chestjournal.org/misc/reprints.shtml).
Correspondence to: Paul D. Thompson, MD, Cardiology Division, Hartford Hospital, 80 Seymour St, Hartford, CT 06102; e-mail: pthomps@harthosp.org
DOI: 10.1378/chest.08-0943

linking statins to ILD, and presents new data on statin-associated ILD from the US Food and Drug Administration adverse event reporting (FDA-AER) database.

Materials and Methods

Literature Review

We performed a systematic literature search for articles published in English and French from 1987 to September 2007 using PubMed and the search terms "HMG-CoA reductase inhibitors" or "statins" in combination with the terms "drug-induced interstitial lung disease," "pulmonary fibrosis," "interstitial pneumonitis," or "parenchymal lung disease." Abstracts and relevant articles were retrieved and reviewed in detail. Selected citations from these articles were also examined. Given the paucity of information, all case reports and case series were included.

Search of Food and Drug Administration Database

Adverse event reports of statin-associated ILD were collected from the Food and Drug Administration database using the electronic search engine (QscanFDA; DrugLogic; Reston, VA) and the search terms "drug-induced interstitial lung disease," "pulmonary fibrosis," "interstitial pneumonitis," or "parenchymal lung disease." All cases of ILD reported to the FDA-AER system as of June 2007 in which a statin was listed as a causative suspect were included in the analysis. The number of reports of statin-associated ILD was determined for each statin. Outcome categories included death and hospitalization without death. We also determined the ratios of ILD reports over all reported adverse events for each statin during the same time period. In addition, we calculated rates of ILD per 1,000,000 prescriptions from 1999 through 2004, a time frame for which we had prescription data available for three commonly prescribed statins: atorvastatin, simvastatin, and pravastatin.

Results

Statin-Induced Parenchymal Lung Damage

The literature search using PubMed yielded eight articles describing a total of 14 case reports of ILD in association with statin use (Table 1). Hill et al[4] in 1995 first described possible statin-induced ILD in a 76-year-old woman accompanying a syndrome of dermatomyositis. The association of statin use with ILD in this report is problematic, however, because

Table 1—*Summary of the 14 Cases Reviewed*

Variables	Hill et al[4] 1995	De Groot et al[6] 1996	Sridhar and Abdulla[8] 1998	Liebhaber et al[5] 1999	Liscoet-Loheac et al[9] 2001	Lantuejoul et al[10] 2002	Veyrac et al[11] 2006
Age, yr/gender	76/female	61/male	67/female	69/female	41/male	51/male	69/female
Statin/daily dose, mg	Simvastatin/10	Simvastatin/10	Fluvastatin/20	Pravastatin/20–40	Pravastatin / none reported	Simvastatin/5; rechallenge: pravastatin	Atorvastatin/ none reported
Time between initiation of statin and onset of symptoms	18 mo	6 mo	1 wk	6 yr	Not reported	6 yr	3 mo
History of lung disease or smoking	Smoking history	Smoking history		Smoking history			Nonsmoker
Chest CT	Extensive bilateral alveolar infiltrates	Peripheral reticular pattern; right pleural effusion; left pleural thickening	Extensive alveolitis and fibrosis	Alveolitis: ground-glass haziness	Bilateral tiny radiopacities; posterior pleural thickening	Ground-glass opacities; focal condensation	Bilateral ground-glass
Lung biopsy	Interstitial inflammation; early fibrosis	Diffuse fibrosis	Alveolar damage; hyaline membranes; intraalveolar organization; atypical reactive pneumocytes	Hypersensitivity pneumonitis		Fibrotic interstitial pneumonia; foamy macrophages	
Outcome after statin stopped	Rash resolved; ventilatory failure; death	Decrease IgE; radiograph: persistent interstitial pattern; decreased DLCO despite prednisone 40 mg/d	Skin improved; respiratory failure and death despite corticosteroids and cyclophosphamide	Prednisone 60 mg/d; CT alveolitis resolved 2 wk after statin stopped	Resolution of symptoms; normalization of radiograph 8 mo after statin stopped	Improved 6 mo after statin stopped; recurrence with pravastatin	Improvement 9 d after statin stopped; persistent infiltrates and reduced DLCO despite 7 mo of corticosteroids

*DLCO = carbon monoxide diffusion capacity.

the patient had very high anti-nuclear antibody titers and tested positive for a human leukocyte antigen serotype DR3, a known associate of rheumatologic diseases, which are recognized causes of ILD. Furthermore, she had a sister who died of idiopathic pulmonary fibrosis. The authors speculated that simvastatin triggered an autoimmune dermatomyositis by producing muscle injury and releasing autoantigens, or by acting as a hapten. We think that it is possible that this was a spontaneous event in a genetically predisposed individual, and not related to statin treatment. Rheumatologic diseases such as systemic lupus erythematosus, mixed connective tissue disorder, antisynthetase syndrome, or systemic scleroderma with lung involvement should be investigated in similar cases. This patient had a smoking history but no reported documentation of prior lung disease.

De Groot et al[6] in 1996 presented a 61-year-old man with ILD and moderate increase in transaminase levels. The patient had an increase in serum IgE levels and eosinophilic pulmonary lavage fluid that suggested an allergic reaction. He had been started on simvastatin 6 months earlier. Eosinophilia in the lavage fluid decreased from 34 to 4%, whereas aspartate aminotransaminase decreased from 740 U/L (normal < 15 U/L) to 100 U/L, and alanine aminotransferase decreased from 930 U/L (normal < 15 U/L) to 43 U/L a few days after discontinuing statin therapy, suggesting a relationship to statin treatment. His pulmonary function test results, although improved, did not normalize, and interstitial infiltrates on CT of the chest persisted despite steroid therapy suggesting permanent fibrotic damage. In this case, high eosinophil content of the bronchoalveolar fluid and high IgE raise the possibility of a hypersensitivity reaction. The temporal correlate of statin use and transaminase increases, one of the most commonly reported side effects of statins,[3] also support the role of statins as a causative agent. This patient had a 31–pack-year smoking history but, again, no reported previous lung disease.

Sridhar and Abdulla[8] in 1998 described a 67-year-old woman with lupus-like syndrome, pulmonary fibrosis, and ARDS. Her symptoms were suggestive of drug-induced lupus and started 1 week after initiating fluvastatin therapy. She also had pruritus, a

Table 1—*Continued*

Walker et al[7] 2007	Walker et al[7] 2007	Walker et al[7] 2007	Walker et al[7] 2007	Walker et al[7] 2007	Walker et al[7] 2007	Walker et al[7] 2007
78/female Atorvastatin/10	78/male Pravastatin/40	74/female Simvastatin/10–20	83/male Simvastatin/40	67/female Simvastatin/40	68/male Simvastatin/10	64/male Atorvastatin/20–40
1 yr	10 yr	3 yr	5 yr	5 yr	2 yr	5 yr
Nonsmoker	COPD and smoking history	COPD and smoking history	Smoking history	Nonsmoker	Smoking history	Smoking history
Extensive bilateral fibrosis	Extensive emphysema; bilateral fibrosis	Diffuse ground-glass appearance	Scattered bilateral fibrosis	Bilateral ground-glass appearance	Bilateral fibrosis	Bilateral ground-glass appearance; fibrosis
	Nonspecific interstitial pneumonitis	Nondiagnostic		Inflammation and fibrosis	Thickened alveoli; interstitial fibrosis	
Continued decline in lung function despite colchicine treatment; persistent dyspnea at 3 yr	Oxygen dependent despite prednisone 50 mg/d; death at 18 mo from respiratory failure	Reduced infiltrate by CT with prednisone 50 mg/d	Continued decline in lung function despite prednisolone 40 mg/d and azathioprine	Improvement in DLCO with after prednisolone	Death after 9 mo despite prednisolone 37.5 mg/d and azathioprine	Small DLCO increase after 2 mo of prednisolone 10 mg/d; stable thereafter

generalized, erythematous rash (with a butterfly distribution on the face), symmetric swelling, and pain in joints of knees and hands. Anti-histone antibodies (usually elevated in drug-induced lupus-like syndrome) and anti-centromere antibodies (usually elevated in scleroderma) were negative. An open lung biopsy was characteristic of drug-induced lung damage. Fluvastatin was stopped after 10 weeks of treatment, but the patient died of respiratory failure. The absence of other causes of ILD and the temporal relationship of the symptoms to fluvastatin therapy suggested statin-induced ILD. No history of lung pathology was documented. Interestingly, steroid therapy improved her skin manifestations but not her respiratory status. This suggests that the statin-induced lung injury might be irreversible in some cases.

Liebhaber et al[5] in 1999 published one case of hypersensitivity pneumonitis associated with statin use. This 69-year-old woman had an erythrocyte sedimentation rate of 101 mm/h. Her angiotensin-converting enzyme, anti-nuclear antibodies, anti-neutrophilic cytoplasmic antibody test, and quantitative Igs were normal. Despite a markedly reduced diffusion capacity and a ground-glass appearance on chest CT, she had a complete resolution of disease 7 weeks after stopping pravastatin and a course of steroid therapy. She was a former smoker, but details of her habit or history of COPD were not documented.

Liscoet-Loheac et al[9] in 2001 reported hypersensitivity pneumonitis in a 41-year-old man treated with pravastatin for an unspecified duration. Peripheral eosinophilia accompanied his presentation, and the chest CT showed multiple diffuse radiopacities with a small pleural effusion. Pravastatin was stopped, his eosinophil count completely normalized, and his chest CT was normal 8 months later. No steroids were used in the management of the patient, which argues against the need of steroids in the treatment of all cases.

Lantuejoul et al[10] in 2002 reported a case of fibrotic nonspecific interstitial pneumonia in a 51-year-old man treated with simvastatin 5 mg/d for 6 years. Lung biopsy revealed airspaces filled with foamy macrophages, a finding classically thought to be characteristic of amiodarone-induced lung toxicity. Ultrastructural analysis of the lung revealed heterogeneous intralysosomal lamellar inclusions in the cytoplasm of the type II pneumocytes, interstitial histiocytes, and endothelial cells, suggesting phospholipidosis, a process common to several drugs that cause lung toxicity. The patient improved with 6 months of steroid therapy but had reappearance of chest infiltrates when treated with pravastatin 1 month later. The patient again improved with discontinuation of this drug. Recurrence of the symptoms after rechallenge highlights the hypothesis of drug-induced lung injury.

Veyrac et al[11] in 2005 described a 69-year-old woman with interstitial pneumopathy. The patient had manifested cutaneous allergic reactions to several generic and brand name statins before her presentation. She had peripheral hypereosinophilia and alveolar eosinophilia associated with the respiratory symptoms. Her symptoms improved dramatically a few days after discontinuation of the medication, but parenchymal lesions on chest CT persisted despite several months of steroid therapy.

Walker et al[7] in 2007 recorded a series of seven cases of interstitial pneumonitis in patients receiving HMG-CoA reductase inhibitors. Five patients were former smokers, and two patients had COPD. Two patients had significantly elevated alveolar eosinophils on the bronchoalveolar fluid analysis, a finding previously described in drug-hypersensitivity pneumonitis. One patient continued statin therapy and died 18 months later from respiratory failure. Five of these patients had a smoking history, and two of them had COPD. In contrast to most previously described cases, these patients did not have extrapulmonary side effects, such as urticaria, dermatomyositis, polymyositis, or lupoid features. This suggests a potential direct lung injury of the drug rather than drug-induced autoimmunity syndromes with subsequent lung involvement.

Table 2—*Reports of ILD Listing a Statin as Suspect and Received by the Food and Drug Administration as of June 2007*

Variables	Lovastatin	Simvastatin	Pravastatin	Atorvastatin	Fluvastatin	Rosuvastatin	Cerivastatin
Reports, No.	14	63	23	47	5	1	9
Age, yr*	65 ± 10	70 ± 9	72 ± 7	69 ± 4	73 ± 9	75	73 ± 3
Male gender, %	33	75	65	74	60	100	100
Death, %	14	43	4	23	0	0	22
Hospitalization†, %	50	46	65	40	40	100	56
ILD/all adverse events rate		0.40	0.29	0.20	0.20	0.01	0.06

*Mean ± SE.

†Hospitalization as outcome without concomitantly reported death. All adverse reports where death was listed as an outcome also listed hospitalization.

The common clinical denominator in most of the cases was an initial presentation of dry cough and dyspnea. When performed, pulmonary function testing showed moderate-to-severe decreases in carbon monoxide diffusion capacity.

The FDA-AER system database contained 162 cases of statin-induced ILD as of June 2007 (Table 2). The total number of reports associated with each drug was as follows: lovastatin, n = 14; simvastatin, n = 63; pravastatin, n = 23; atorvastatin, n = 47; fluvastatin, n = 5; rosuvastatin, n = 1; and cerivastatin, n = 9. The subjects were predominantly men (72%) with a mean age of 71 years. Since the 162 cases represent the absolute number of cases retrieved from the complete data set, it is difficult to predict the prevalence of statin-induced ILD in the general population. We calculated ILD report rates per 1,000,000 prescriptions for atorvastatin (0.07 per 1,000,000 prescriptions), simvastatin (0.18 per 1,000,000 prescriptions), and pravastatin (0.18 per 1,000,000 prescriptions). When the ratio of ILD reports to total adverse events reported were calculated for the various statins, we observed that for every 10,000 reports of a statin-associated adverse event, approximately 1 to 40 reports were for an ILD, with the lowest ratio observed with rosuvastatin (1 in 10,000) and the highest with simvastatin (4 in 1,000) [Table 2].

Discussion

Definition of ILD

ILD, also known as *diffuse parenchymal lung disease*, comprises a diverse group of disorders that may share similar pathologic findings. They may have cellular infiltration, scarring, and/or architectural disruption of the pulmonary parenchyma involving alveolar lining cells, small and large airways, endothelial basement membranes, and occasionally the pleura.[12]

Drug-Induced ILD

The lungs are vulnerable targets for many drugs. More than 150 pharmacologic agents have been reported to cause adverse pulmonary reactions.[13,14] Typically, drug-induced ILD occurs with an acute (days to weeks) or subacute (months) presentation. Commonly patients complain of nonspecific symptoms such as dyspnea on exertion and a nonproductive cough. Classically, a temporal relationship between the administration of the drug and the development of symptoms exists. Uncommonly, the lung disease appears weeks to years after the drug has been discontinued.[15] The symptoms and injury usually remit with cessation of the drug and can recur with rechallenge. Chest radiograph, high-resolution CT, and even histologic findings are often nondiagnostic. Nevertheless, various pulmonary patterns of response have been described in drug-induced pulmonary disease.[16] The most common abnormality on high-resolution CT is reticular or ground-glass opacities with or without consolidation.

Drug-induced ILD can mimic other ILDs and is considered a condition of exclusion rather than a specific entity.[14] Therefore, when suspecting statin-induced ILD, all other explanations for the lung injury must be excluded and no other drugs can account for the reaction.

Proposed Mechanism of Damage

The mechanism of injury in possible statin-induced ILD, as in most causes of ILD, is unknown. The time to the onset of symptoms can vary from months to years after initiation of statin therapy.[5] Some authors[10] have proposed that the potential toxic effect is mediated by the inhibition of phospholipases as part of the statin effect on lipid metabolism. This could produce phospholipid-rich intralysosomal lamellar inclusions in type II pneumocytes (the cell type normally producing surfactant), histiocytes, and endothelial cells, causing an amphiphilic drug-like toxic lung injury.[10] Some anorectic drugs, neuroleptics, antidepressants, and amiodarone are thought to produce lung injury via this mechanism.[17,18] Electron microscopy of lung biopsy specimens from patients with amiodarone pulmonary toxicity characteristically reveals an abundance of similar granular and lamellar lysosomal inclusion bodies.

Amiodarone and HMG-CoA reductase inhibitors share an amphiphilic structure. They contain an apolar ring and a hydrophilic cationic side chain. It is believed that these structural characteristics may be related to the development of drug-induced phospholipidosis.[19,20]

The compounds known to induce lipidosis-like changes display a distinctive pharmacokinetic pattern in which the tissue/plasma drug ratios rise with the duration of the treatment.[18] This could explain why possible statin-induced ILD may present after months or even years of treatment.

More recently, the disruption of oxygen consumption, mitochondrial function, and adenosine triphosphate levels caused by amiodarone have been hypothesized as potential triggers of amiodarone lung cytotoxicity.[21] Statins may also impact mitochondrial metabolism. The potential of statins to interfere with energy metabolism in the muscle mitochondria by disrupting the electron transport chain has been postulated as a potential explanation for statin-

induced myopathy.[22,23] Simultaneous use of these medications can possibly magnify mitochondrial toxicity.

Another potential mechanism for statin-induced lung injury is immune mediated, in which oxygen free radicals cause cell injury and inflammation. Intracellular antigens could trigger an immunologic response with cytotoxic antibodies, immune complexes, cytotoxic T lymphocytes, and CD4+ T cells producing cytokines.[24] This mechanism would explain the presence of positive autoantibodies and systemic reversible clinical features consistent with the polymyositis,[5,25] dermatomyositis,[4,26,27] and drug-induced lupus[8,28] present in some published reports. In general, in cases of drug-induced autoimmunity, withdrawal of the offending agent produces rapid clinical improvement,[24] as noted in some cases.[5,9,10] However, it is disconcerting that antibodies or immune complexes have not been identified in any of the lung biopsy findings.

It is also not clear why the injury occurs in so few of the millions of patients treated with statins. This suggests that the lung reaction requires some genetic or other predisposing factors or that the association of ILD with statins is by chance alone. The lung injury seems to be class specific because ILD has been reported with pravastatin,[9,26] simvastatin,[4,10,29] fluvastatin,[8] and atorvastatin.[7,11]

Therapeutic Options

Discontinuing the statin improves and may completely resolve, symptoms.[5,9] Most, but not all, patients respond to steroids or immunosuppressive agents.[11] Partial responses to steroids and variable degrees of reversibility after discontinuation of the drug may imply later stages of lung injury, whereas complete clinical and paraclinical resolution may reflect early disease.

Potential Therapeutic Use of Statins in Respiratory Disease

Paradoxically, there also are some studies[30–34] that evaluate the potential beneficial effect of statins in ILD and pulmonary fibrosis. These studies have been based on the ability of statins to modulate local fibroproliferative pathways in experimental models. Nadrous et al[34] retrospectively compared survival of patients with idiopathic pulmonary fibrosis treated with angiotensin-converting enzyme inhibitors and/or statins with the survival of a cohort of patients with IPF not receiving these drugs. They did not find significant difference in survival. Alexeeff et al,[33] evaluated the rate of lung function decline in elderly men from the Normative Aging Study. Those not using statins had higher rates of lung function decline than those using statins (23.9 mL/yr vs 10.9 mL/yr [$p < 0.001$ for all], respectively). Even though this study did not specifically include patients with ILD, these results indicate a potential beneficial effect of statins in lung function in general.

The objective of our study is not to dispute the potential benefit of statins in the treatment of non–statin-induced ILD, but rather to raise awareness about the potential pulmonary side effect that this class of medications might have in a very small subgroup of patients. Advances in the field of pharmacogenetics might ultimately help identify these susceptible individuals.

Clinical Implications

The evidence linking statins to lung injury is not definitive. These cases of ILD reported as being due to statins could simply reflect background prevalence of ILD and that statins are not the cause. Such individuals could have had idiopathic ILD inappropriately attributed to the statin. Nevertheless, such an association is possible, and clinicians should consider this possibility in patients on statins who present with ILD. We can make no definitive recommendations for treatment given the relatively small number of cases. However, it is prudent to stop the statin in any patient who has unexplained pulmonary symptoms and radiographic changes. The contribution of steroid therapy in the treatment of statin-induced ILD is uncertain, but based on anecdotal experience[6,10] steroid therapy should be attempted when statin withdrawal fails to improve symptoms.

Summary

Statin-induced ILD has emerged as a possible new side effect of statin therapy. ILD has been reported with most statins, suggesting that statin-induced ILD is a class effect and not a specific statin effect. The mechanism of injury remains controversial. Because of the paucity of data and novelty of these findings, it is difficult to extrapolate a definite degree of association between these drugs and ILD. It is possible that future studies will identify a more distinct pattern of occurrence, a precise mechanism of disease, and a degree of causal relationship, as more physicians become aware of this potential side effect.

References

1 Rosenson RS, Tangney CC. Antiatherothrombotic properties of statins: implications for cardiovascular event reduction. JAMA 1998; 279:1643–1650

2 Bellosta S, Paoletti R, Corsini A. Safety of statins: focus on

clinical pharmacokinetics and drug interactions. Circulation 2004; 109(23 suppl 1):III50–III57
3 Alsheikh-Ali AA, Ambrose MS, Kuvin JT, et al. The safety of rosuvastatin as used in common clinical practice: a postmarketing analysis. Circulation 2005; 111:3051–3057
4 Hill C, Zeitz C, Kirkham B. Dermatomyositis with lung involvement in a patient treated with simvastatin. Aust N Z J Med 1995; 25:745–746
5 Liebhaber MI, Wright RS, Gelberg HJ, et al. Polymyalgia, hypersensitivity pneumonitis and other reactions in patients receiving HMG-CoA reductase inhibitors: a report of ten cases. Chest 1999; 115:886–889
6 De Groot RE, Willems LN, Dijkman JH. Interstitial lung disease with pleural effusion caused by simvastin. J Intern Med 1996; 239:361–363
7 Walker T, McCaffery J, Steinfort C. Potential link between HMG-CoA reductase inhibitor (statin) use and interstitial lung disease. Med J Aust 2007; 186:91–94
8 Sridhar MK, Abdulla A. Fatal lupus-like syndrome and ARDS induced by fluvastatin [letter]. Lancet 1998; 352:114
9 Liscoet-Loheac N, Andre N, Couturaud F, et al. Hypersensitivity pneumonitis in a patient taking pravastatin [in French]. Rev Mal Respir 2001; 18:426–428
10 Lantuejoul S, Brambilla E, Brambilla C, et al. Statin-induced fibrotic nonspecific interstitial pneumonia. Eur Respir J 2002; 19:577–580
11 Veyrac G, Cellerin L, Jolliet P. Un cas de pneumopathie interstitielle rapporte sous atorvastatine (Tahor): synthese des cas publies sous statines. Therapie 2006; 61:57–67
12 Chesnutt M, Murray J, Prendergast P. Interstitial lung disease In: Tierney LM, McPhee SJ, Papadakis MA, eds. Current medical diagnosis and treatment. 46th ed. Available at: http://www.accessmedicine.com. Accessed September 22, 2007
13 Foucher P, Philippe C. The drug-induced lung diseases. Available at: http://pneumotox.com. Accessed November 20, 2007
14 Limper AH, Rosenow EC III. Drug-induced interstitial lung disease. Curr Opin Pulm Med 1996; 2:396–404
15 Limper A. Drug induced pulmonary disease In: Mason RJ, Murray JF, Broaddus VC, eds. Murray and Nadel's textbook of respiratory medicine. 4th ed. Available at: http://www.mdconsult.com. Accessed September 22, 2007
16 Demeter SL, Ahmad M, Tomashefski JF. Drug-induced pulmonary disease: part I. Patterns of response. Cleve Clin Quar 1979; 46:89–99
17 Rossi SE, Erasmus JJ, McAdams HP, et al. Pulmonary drug toxicity: radiologic and pathologic manifestations. Radiographics 2000; 20:1245–1259
18 Dake MD, Madison JM, Montgomery CK, et al. Electron microscopic demonstration of lysosomal inclusion bodies in lung, liver, lymph nodes, and blood leukocytes of patients with amiodarone pulmonary toxicity. Am J Med 1985; 78: 506–512
19 Lullmann H, Lullmann-Rauch R, Wassermann O. Drug-induced phospholipidoses: II. Tissue distribution of the amphiphilic drug chlorphentermine. CRC Crit Rev Toxicol 1975; 4:185–218
20 Schmien R, Seiler KU, Wassermann O. Drug-induced phospholipidosis: I. Lipid composition and chlorphentermine content of rat lung tissue and alveolar macrophages after chronic treatment. Naunyn-Schmiedeberg's Arch Pharmacol 1974; 283:331–334
21 Bolt MW, Card JW, Racz WJ, et al. Disruption of mitochondrial function and cellular ATP levels by amiodarone and N-desethylamiodarone in initiation of amiodarone-induced pulmonary cytotoxicity. J Pharmacol Exp Ther 2001; 298: 1280–1289
22 Folkers K, Langsjoen P, Willis R, et al. Lovastatin decreases coenzyme Q levels in humans. Proc Natl Acad Sci U S A 1990; 87:8931–8934
23 Willis RA, Folkers K, Tucker JL, et al. Lovastatin decreases coenzyme Q levels in rats. Proc Natl Acad Sci U S A 1990; 87:8928–8930
24 Bigazzi PE. Autoimmunity caused by xenobiotics. Toxicology 1997; 119:1–21
25 Fauchais AL, Iba Ba J, Maurage P, et al. Polymyositis induced or associated with lipid-lowering drugs: five cases [in French]. Rev Med Interne 2004; 25:294–298
26 Zuech P, Pauwels C, Duthoit C, et al. Pravastatin-induced dermatomyositis [in French]. Rev Med Interne 2005; 26:897–902
27 Dourmishev AL, Dourmishev LA. Dermatomyositis and drugs. Adv Exp Med Biol 1999; 455:187–191
28 Ahmad S. Lovastatin-induced lupus erythematosus. Arch Intern Med 1991; 151:1667–1668
29 Yoshioka S, Mukae H, Ishii H, et al. A case of drug-induced pneumonia possibly associated with simvastatin [in Japanese]. Nihon Kokyuki Gakkai Zasshi 2005; 43:600–604
30 Watts KL, Sampson EM, Schultz GS, et al. Simvastatin inhibits growth factor expression and modulates profibrogenic markers in lung fibroblasts. Am J Respir Cell Mol Biol 2005; 32:290–300
31 Watts KL, Cottrell E, Hoban PR, et al. RhoA signaling modulates cyclin D1 expression in human lung fibroblasts; implications for idiopathic pulmonary fibrosis. Respir Res 2006; 7:1–14
32 Tan A, Levrey H, Dahm C, et al. Lovastatin induces fibroblast apoptosis *in vitro* and *in vivo*: a possible therapy for fibroproliferative disorders. Am J Respir Crit Care Med 1999; 159:220–227
33 Alexeeff SE, Litonjua AA, Sparrow D, et al. Statin use reduces decline in lung function: VA Normatice Aging Study. Am J Respir Crit Care Med 2007; 176:742–747
34 Nadrous HF, Ryu JH, Douglas WW, et al. Impact of angiotensin-converting enzyme inhibitors and statins on survival in idiopathic pulmonary fibrosis. Chest 2004; 126:438–446